

KILLARA HIGH SCHOOL

HSC ASSESSMENT 1 - 2006

YEAR 12 - MATHEMATICS

Time allowed: 1 hour

Instructions:

Attempt all questions

Approved calculators may be used

Start your answer to each question on a new page.

Total marks: 50

Question 1 (31 marks): Calculus

Marks

a. Differentiate the following functions.

i. $y = 2x - 5x^2 + 9$ 1

ii. $y = 3\sqrt{x} - x^4 + \frac{2}{x^3} + 7$ 2

iii. $y = (2x^5 - 1)(6 + x)$ 2

iv. $y = \frac{3x^5 - 10}{1 + x^3}$ 2

v. $y = (3x^4 - 5x)^6$ 1

vi. $y = \frac{8}{\sqrt{3x^5 + 1}}$ 1

b. Find $\lim_{x \rightarrow \infty} \frac{x^2 - 4x + 1}{x^2 - 3}$ 2

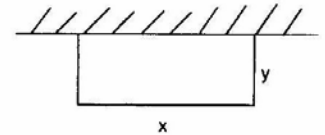
- c. i. Find the coordinates of the point P where the tangent to the curve $y = x^2 - 4x$ is parallel to the line $y = 2x + 1$. 2
- ii. Find the equation of the normal at the point P. 2
- iii. Find the coordinates of the points M and N where the normal cuts the X and Y axes respectively. 2
- iv. Find the area of $\triangle OMN$, where O is the origin. 1

c. Given that $y = 3x^2 - x^3$,

- i. Find the stationary points and distinguish between them. 3
- ii. Find the point of inflexion. 2
- iii. Sketch the curve. 3

- d. Farmer Brown has 500 m of wire with which to build a rectangular paddock. To save wire he decides to use an existing fence as one side of the paddock.

- i. Show that the area, $A \text{ m}^2$, of the paddock is given by $A = \frac{(500 - x)x}{2}$ 2



- ii. Find the dimensions of the paddock which would enclose the greatest area. 3

Question 2 (19 marks): Sequence and series

- a. The first term of an arithmetic progression is -25. The common difference is 4.
- i. Write down the next two terms. 1
 - ii. Find the 25th term. 1
- b. Find $\sum_{n=9}^{20} 2n - 1$ 2
- c. What is the least number of terms of the series $1 + 3 + 9 + \dots$ that need to be taken to give a sum greater than 300. 3
- d. The 3rd term of a geometric series is 24 and the 6th term is 1536. Find
- i. Find the common ratio. 2
 - ii. Find the first term. 1
- e. By expressing the recurring decimal $0.12\dot{3}$ as the sum of an infinite number of terms of a series, find its value as a fraction in its simplest form. 3
- f. A woman borrows \$350000 in order to buy a house. Interest of 8.4% p.a. on the loan is calculated monthly on the balance still owing. The equal repayments, of \$M each, are made monthly and the loan is to be repaid over 25 years.
- i. Show that the amount A_2 still owing after making two repayments of \$M each is given by $A_2 = 350000(1.007)^2 - M(1 + 1.007)$ 2
 - ii. Find the amount, A, owing at the end of 25 years, in term of M. 1
 - iii. Hence, show that $M = \frac{350000(1.007)^{300}(0.007)}{1.007^{300} - 1}$ and find the value of M to the nearest dollar. 3

End of paper

Solution

Question 1: 21 marks

3) (i) $y' = 2 - 10x$ (1)

(ii) $y' = \frac{3}{2\sqrt{x}} - 4x^3 - \frac{6}{x^4}$ (2)

(iii) $y' = 10x^4(6+x) + 2x^5 - 1$ (1)
 $= 60x^4 + 10x^5 + 2x^5 - 1$
 $= 12x^5 + 60x^4 - 1$ (1)

iv) $y' = \frac{15x^4(1+x^3) - 3x^2(3x^5-10)}{(1+x^3)^2}$ (1)
 $= \frac{15x^4 + 15x^7 - 9x^7 + 30x^2}{(1+x^3)^2}$
 $= \frac{6x^7 + 15x^4 + 30x^2}{(1+x^3)^2}$ (1)

v) $y' = 6(3x^4 - 5x)^5 \times (12x^3 - 5)$ (1)
 $= 6(12x^3 - 5)(3x^4 - 5x)^5$

vi) $y' = \frac{-4}{\sqrt{(3x^5+1)^3}} \times 15x^4$
 $= \frac{-60x^4}{\sqrt{(3x^5+1)^3}}$ (1)

(b) $= \lim_{x \rightarrow \infty} \frac{\frac{x}{x^2} - \frac{4x}{x^2} + \frac{1}{x^2}}{\frac{x^2}{x^2} - \frac{3}{x^2}} = 1$ (2)

(c) (i) $y' = 2x - 4$ (Sub. $m=2$)

$2 = 2x - 4$ (1)

$\begin{cases} x = 3 \\ y = -3 \end{cases}$ (1)

(ii) $m_{\text{normal}} = -\frac{1}{2}$ (1)

$\frac{y+3}{x-3} = -\frac{1}{2} \Rightarrow 2y+6 = -x+3$

$2y+x+3 = 0$ (1)

(iii) $\therefore M(0, -\frac{3}{2})$ (1) and $N(-3, 0)$ (1)

iv) $A = \frac{1}{2} \times 3 \times \frac{3}{2} = \frac{9}{4} = 2.25 \text{ sq. units}$ (1)

(d) $y' = 6x - 3x^2$

(i) stationary points: $\begin{cases} 6x - 3x^2 = 0 & x=0, y=0 \\ 3x(2-x) = 0 & x=2, y=4 \end{cases}$

$y'' = 6 - 6x$ $f''(0) > 0$

min. pt.
(0, 0)

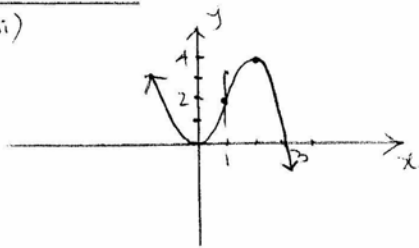
$f''(2) < 0$

max. point. (1)
(2, 4)

(ii) Pts. of inflexion: $6 - 6x = 0 \Rightarrow x = 1$
 $f''(x < 1) > 0$ $f''(x > 1) < 0$ (1) $\therefore$ pt. of inflexion (1, 2) (1)

Q1 continue

(c)(iii)



① correct shape

② showing all critical points

③ any missing or incorrect

(d)(i)

$$x + 2y = 500$$

$$y = \frac{500 - x}{2}$$

①

$$A = \frac{500 - x}{2} \times x = \frac{(500 - x)x}{2} \quad (1)$$

$$(ii) \frac{dA}{dx} = 250 - x = 0$$

$$\Rightarrow x = 250 \quad (1)$$

$$A''(x) = -1 < 0$$

$\therefore$ always concave down

$\therefore x = 250$ is a max (1)

$$y = 125 \quad (1)$$

Quest. 2 : 17 marks

(a)

$$(i) -21, -17 \quad (1)$$

$$(ii) T_{25} = -25 + (25 - 4) \times 4 = 71 \quad (1)$$

(b)

$$17, 19, \dots, 39 \quad (1)$$

$$S_{12} = \frac{12}{2} (17 + 39) = 336 \quad (1)$$

(c)

$$\frac{1(3^n - 1)}{3 - 1} > 300 \quad (1) \Rightarrow 3^n - 1 > 600$$

$$n > \frac{\log 601}{\log 3} \quad (1)$$

$$n > 5.822$$

$$\therefore n = 6 \quad (1)$$

(d)

$$(i) ar^5 = 1536 \quad (1)$$

$$ar^2 = 24 \quad (1)$$

$$\frac{ar^5}{ar^2} = \frac{1536}{24} \Rightarrow r^3 = 64 \Rightarrow r = 4 \quad (1)$$

(ii)

$$24 = 16a \Rightarrow a = 1.5 \quad (1)$$

(e)

$$0.12\dot{3} = 0.12333\dots = \frac{12}{10} + \frac{3}{1000} + \frac{3}{10000} + \dots \quad (1)$$

$$S_{\infty} = \frac{\frac{3}{1000}}{1 - \frac{1}{10}} = \frac{1}{300} \quad (1)$$

$$\therefore 0.12\dot{3} = \frac{12}{10} + \frac{1}{300} = \frac{37}{300} \quad (1)$$

(e)

$$i) A_1 = 350000 (1 + .007)^1$$

$$= 350,000 (1.007) - M$$

$$A_2 = [350,000 (1.007) - M] 1.007 - M \quad (1)$$

$$= 350000 (1.007)^2 - M (1 + 1.007) \quad (1)$$

$$\vdots$$

$$(ii) A_{25} = 350000 (1.007)^{300} - M (1 + 1.007 + \dots + 1.007^{299}) \quad (1)$$

(iii)

At end of 25 years $A_{25} = 0$

$$0 = 350000 (1.007)^{300} - M (1 + 1.007 + \dots + 1.007^{299})$$

$$S_{300} = \frac{1(1.007^{300} - 1)}{.007}$$

$$\Rightarrow 0 = 350000 (1.007)^{300} - \frac{M (1.007^{300} - 1)}{.007} \quad (1)$$

$$\frac{M (1.007^{300} - 1)}{.007} = 350000 (1.007)^{300}$$

$$\therefore M = \frac{350000 (1.007)^{300} (.007)}{1.007^{300} - 1} \quad (1)$$

$$\boxed{M = \$2795} \quad (1)$$

$$r = 8.4\% \div 12$$

$$= .007$$

$$n = 12 \times 25$$

$$= 300 \text{ months.}$$